

PS5 Linux Cheatsheet

Jailbroken PS5 (7.20) · ps5-linux CachyOS Gaming Image · Full Setup
Reference

Sessions · Controllers · Bluetooth · Troubleshooting

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1. System Map

Component	What it is
Gaming mode	gamescope session + Steam Big Picture, auto-starts on tty1 at boot (default)
Desktop mode	KDE Plasma on X11 (startx + startplasma-x11 + kwin-x11)
Session token	<code>~/.config/ps5-next-session</code> — contains <code>gamescope</code> or <code>plasmamax11</code> ; read at every tty1 login by <code>/etc/profile.d/ps5-tty-session.sh</code>
Boot default service	<code>ps5-session-default.service</code> resets the token to <code>gamescope</code> at every boot → boot always lands in gaming mode
User / host	user <code>steam</code> (image default), hostname per install, kernel 7.1.x (custom ps5-linux; tested 7.1.3–7.1.4)
Network	Ethernet <code>enp64s0f2</code> , Wi-Fi <code>wlp64s0f7</code> (Marvell, mwifiex driver)
Internal Bluetooth	Working (Marvell 1286:2059 via xhci) since script v10: udev rule keeps btusb off phantom interface 5-1:1.2 + kernel HID path. Upstream fix: ps5-linux-patches PR #19 (2026-06-23); pairing policy fix (JustWorksRepairing) in script v11

2. Session Switching

Direction	Method	Notes
Boot → Gaming	automatic	boot service resets token to gamescope every boot
Gaming → Desktop	"Switch to Desktop" tile in Big Picture (non-Steam shortcut)	soft switch, ~15 s, kills zombie sessions first; no watchdog
Desktop → Gaming	Steam's own "Switch to Gaming Mode" button (works)	alternatively: <code>~/rescue-gaming.sh</code> via SSH
Emergency	<code>~/rescue-gaming.sh</code> from SSH	fixes black screen after a failed desktop switch; worst case: hold power button — boot lands in gaming anyway

Files behind the tile

File	Purpose
<code>~/switch-to-desktop.sh (v7)</code>	writes <code>plasmamax11</code> token, kills <code>kwin_wayland/plasmashell/Xorg/gamescope</code> , restarts <code>getty@tty1</code>
<code>~/.local/share/applications/switch-to-desktop.desktop</code>	makes the tile appear in Steam's "Add Non-Steam Game" list
<code>~/rescue-gaming.sh</code>	writes <code>gamescope</code> token + restarts <code>getty@tty1</code>
<code>/etc/systemd/system/ps5-session-default.service</code>	enabled; resets token at boot
<code>/usr/bin/steamos-session-select</code>	patched: writes <code>plasmamax11</code> (not <code>plasma</code>); backup at <code>.bak</code> . Note: Steam's UI button never calls it — that's why the tile exists

Known quirks. The DP → HDMI bridge chip occasionally black-screens a soft gamescope → Plasma handoff (intermittent, hardware-level, documented by the image authors). If the desktop doesn't appear: `~/rescue-gaming.sh` via SSH, then just try the tile again. Steam's own "Switch to Desktop" menu entry does NOT work on this image — it never calls the session script; use the tile.

3. SSH & Network

SSH from Windows PowerShell

```
ssh steam@<ip>
```

Find the IP on the console itself (or router client list):

```
ip -o -4 addr show scope global | awk '{print $2, $4}'
```

SSH timeouts on all known IPs? The console is probably sitting at the jailbreak loader, or the DHCP lease changed. Check the loader screen / router.

"REMOTE HOST IDENTIFICATION HAS CHANGED" after a re-flash? Expected — a fresh install generates new SSH host keys while your PC remembers the old ones. Fix once: `ssh-keygen -R <ip>`, then reconnect and accept the new fingerprint.

Wi-Fi

```
sudo nmcli device wifi list
sudo nmcli device wifi connect "SSID with spaces" password "PASSWORD"
nmcli device status
```

- SSIDs with spaces **must** be quoted; underscores do not work.
- "Not authorized to control networking" over SSH = polkit → use `sudo`.
- Wi-Fi set up via `sudo nmcli` becomes a **system connection**: auto-connects at boot and never triggers KDE Wallet prompts (the desktop applet stores secrets in KWallet and nags; nmcli does not).

Pasting into SSH (important!)

PowerShell multi-line pastes race the terminal and corrupt files. Rules that saved us repeatedly:

- Paste one block at a time; wait for the prompt before the next block.
- For files: chunked heredocs (~70 lines max per chunk) with `wc -l` checkpoints.
- **Always verify:** `sha256sum file` against the expected hash. Hash `e3b0c44298fc...b855` = empty file = paste got eaten.
- Never paste bash heredocs into a bare PowerShell prompt — check for `[steam@ps5linux ~]$` first.

4. Controllers

DualSense via internal Bluetooth (primary)

- Pair: the **Pair DualSense** desktop/Steam tile (`~/pair-dualsense-tile.sh`, no SSH needed — the pad works wired via USB meanwhile) or `~/pair-dualsense.sh / ~/ps5-final.sh pair` over SSH — paperclip reset, PS + Create, ~40 s.
- Works because of: udev rule `99-ps5-bt-no-phantom.rules` (blocks the phantom hci1 on interface 5-1:1.2), `UserspaceHID=false` in `/etc/bluetooth/input.conf`, `hid-playstation` module, `JustWorksRepairing=always` + `PairableTimeout=0` in `/etc/bluetooth/main.conf`, pairable **set-and-verified** before every `pair` (v23+), NO ERTM-off hack.
- Sanity check: `ls /sys/class/bluetooth/` must show ONLY `hci0`; `/proc/bus/input/devices` lists the DualSense when connected.

DualSense via 8BitDo USB Wireless Adapter v1 (backup)

- Grey adapter, firmware updated to **v2.07** with the "Upgrade tool" from support.8bitdo.com/firmware-updater.html (the old 2020 updater is obsolete — that was the whole problem).
- DualSense support needs firmware ≥ 2.0 .
- Pairing: adapter pair button + **PS** + **Create** on the DualSense.
- Presents as "Xbox 360 Controller" (XInput) — pure USB HID, zero Bluetooth involved.

Gamepad as mouse on the desktop

- Steam autostarts minimized on the desktop: `~/config/autostart/steam.desktop` (`steam -silent`).
- Layout: Steam → Einstellungen → Controller → Erweiterte Einstellungen → **Spielfremde Controller-Layouts** → **Desktop-Layout**. Right stick = mouse, A = left click, B = right click.
- Applies **only** on the desktop (no game / no Big Picture active) — gaming controls are untouched.
- **Per controller type**: the layout is stored per pad — set it separately for the DualSense (internal BT) and the Xbox 360 (8BitDo). PS default maps the touchpad as mouse; add right stick or gyro manually.

Bluetooth dongles (CSR8510: CSL + LogiLink)

BT 4.0 dongles, kept as spares. Internal BT works since the udev-rule + pairing-policy fixes; the 8BitDo adapter remains a solid backup (zero Bluetooth stack involved).

5. Big Picture: Adding Games & Artwork

Non-Steam games (and the two tiles!)

1. Place a `.desktop` file in `~/.local/share/applications/` (example: `firefox.desktop`).
2. Steam (desktop mode) → Games → Add a Non-Steam Game → pick it.
3. **The two tiles** ("Switch to Desktop", "Pair DualSense") only appear in Big Picture after this one-time add — get to the desktop via `~/switch-to-desktop.sh` over SSH, add both in Steam, return via Steam's "Switch to Gaming Mode" button.
4. Tile artwork: matching capsule/hero/logo/icon sets ship in the repo's `artwork/` folder; set them per tile via Steam → right-click → Manage → Set custom artwork/hero/logo.

Artwork via Steam ROM Manager (SRM)

- AppImage needs `fuse2` installed (or run with `--appimage-extract-and-run`).
- Pulls artwork from SteamGridDB; the "Non-SRM Shortcuts" parser adds artwork to existing shortcuts.
- SteamGridDB dimensions: Hauptkapsel 600×900, Hintergrund 3840×1240, Logo 1280×720, Icon 128×128.

6. Master Script (ps5-final.sh)

- Lives at `~/ps5-final.sh`; run after every re-flash: `bash ~/ps5-final.sh`
- Standalone pairing mode: `bash ~/ps5-final.sh pair` (MAC auto-detection, option to keep existing controllers)
- Steps: internet check · keymap · printk · BT fixes (udev phantom rule, kernel HID) · SSH · BlueZ (UserspaceHID=false) · mwiflex (kernel-aware) · ps5-linux-tools · helpers · optional 1080p cmdline flag (capture-card fallback-EDID fix; toggle via `~/toggle-1080p.sh`) · screensaver off · Firefox · desktop hardening (plasma-x11-session, kwin-x11) · session machinery bundle (tile, rescue script, boot service, session-select patch, Steam autostart, deps)
- Transfer to the PS5 only via chunked heredocs + `sha256sum` verification (see §3).

7. Troubleshooting

Symptom	Cause	Fix
Black screen after tile switch	DP → HDMI bridge quirk (intermittent)	<code>~/rescue-gaming.sh</code> via SSH; retry tile
Black screen WITH movable cursor	zombie Wayland session squatting on GPU	<code>sudo loginctl terminate-user steam</code> (kills SSH too); v7 tile prevents it
Boots into desktop every time	token stuck on plasmax11	fixed by <code>ps5-session-default.service</code> ; check <code>systemctl status ps5-session-default</code>
No title bars, Alt+F4/Tab dead	missing X11 window manager (KDE packaging split)	<code>sudo pacman -S plasma-x11-session kwin-x11</code> (in master script)
pacman 404 on a package	stale mirror DB	<code>sudo pacman -Sy</code> first, then retry
Package target not found (whiptail)	Arch ships it in <code>libnewt</code>	install <code>libnewt</code>
AppImage won't start	FUSE missing	<code>sudo pacman -S fuse2</code> or <code>--appimage-extract-and-run</code>
nmcli "Not authorized"	polkit over SSH	use <code>sudo nmcli</code>
nmcli "invalid extra argument"	SSID spaces	quote the SSID
Wi-Fi connect fails: "Secrets were required, but not provided" / "No agents were available"	an earlier GUI attempt (Big Picture/applet) left a KWallet-backed profile; sudo has no secret agent	<code>sudo nmcli connection delete "SSID"</code> , then reconnect via <code>sudo nmcli device wifi connect ...</code> (script does this automatically)
Desktop nags for password on Wi-Fi scans/mounts/admin actions	polkit demands auth for GUI admin actions on this image	fixed by <code>/etc/polkit-1/rules.d/50-steam-console-admin.rules</code> (script step 14; single-user consoles only)
"KDE Wallet" asks to be unlocked	secret vault has nothing to guard once Wi-Fi is a system connection	disabled by the script (<code>kwalletrc Wallet/Enabled=false</code>); re-enable via System Settings → KDE Wallet
New app won't start: "version 'GLIBC_2.xx' not found"	partial upgrade: new package on old rolling-release base	<code>sudo pacman -Syu</code> (full upgrade), reboot; script step 2 offers this
SSH session hung after UI "Switch to Desktop"	Steam button kills the tty session (by design, button is useless)	use the tile instead
DualSense won't pair at all (internal BT)	phantom hci1: btusb double-bound interface 5-1:1.2	udev rule <code>99-ps5-bt-no-phantom.rules</code> (script v10+); fixed upstream PR #19 (kernels after 2026-06-23); verify <code>ls /sys/class/bluetooth/</code>

DualSense pairs but drops ~2 s later ("terminated by local host")	ERTM-off relic and/or UserspaceHID=true	delete /etc/modprobe.d/bluetooth-ertm.conf, set UserspaceHID=false, restart bluetooth
"Pairing successful" then drop + bond deletes itself	DualSense re-pairs ~1.5 s after pairing to upgrade its link key ("repairing"); BlueZ rejects that round if the adapter left pairable mode or the repairing policy says no	JustWorksRepairing=always in /etc/bluetooth/main.conf + keep adapter pairable during pairing (helper does both). The drop right after pairing is NORMAL — the stored key works from then on
bluetoothctl connect / KDE Connect button fail with br- connection-create- socket	NOT a bug — an idle DualSense ignores host-initiated connects; only the pad can reconnect (packet-verified)	tap PS once — the pad connects itself. First tap after a boot may drop once (kernel hidp module loading); script preloads hidp at boot
DualSense connected, solid light bar, but no input	no HID device (userspace uhid path broken)	UserspaceHID=false + reconnect; check /proc/bus/input/devices
Pad returns to previous host after pairing	link never established; pad keeps last- good host	fix the drop first; unplug the 8BitDo adapter while testing
Screen frozen on console log at boot	quiet-printk is by design; system is up	SSH in; check capture card/OBS
OBS/preview black, but a directly attached TV works	capture card's fallback EDID (nothing on its OUT port) advertised modes the capture side can't encode	enable the 1080p flag: ~/toggle- 1080p.sh + reboot (revert the same way)
Script file corrupted after paste	PowerShell paste race	chunked heredocs + sha256 verification (§3)
"Pair DualSense" tile runs but shows no pop- ups in Big Picture	kdiallog --passivepopup needs Plasma's notification infra — absent in the gamescope gaming session (script ≤v13)	v14+ tile uses timed zenity dialogs instead; progress also in /tmp/pair- tile.log — follow the light bar: double-blink = pairing, solid = connected
"Pairing successful", then bond vanishes at the drop (Paired: no)	host-initiated bond is TEMPORARY: BlueZ answers SSP with "No Bonding" and sends MGMT Unpair Device at the disconnect (btmon-verified). The permanent bond is pad-made	not a failure — tap PS ; the pad re- pairs itself (General Bonding) and connects. v16+ scripts wait for this instead of bailing; a still-connected pad must first be turned off (hold PS ~10 s)
Pad connects after pairing but stays Paired: no, no input device	race: pairable was closed when the pad CONNECTED, but its re-pair request arrives ~2 s later → "Pairing Not Allowed (0x18)"; unbonded link has no encrypted HID (btmon-verified)	v17+ keeps pairable open until Paired: yes (50 s window). On older scripts: re-run pairing; tap PS again if the light goes dark before solid

SSH suddenly "Permission denied" though the password is right	too many mistyped attempts → <code>pam_faillock</code> locks the account temporarily (nothing in the setup touches passwords)	on the console desktop: <code>pkexec faillock --user steam --reset</code> , if needed <code>pkexec passwd steam</code> (polkit rule → no old password needed); check <code>journalctl -u sshd -n 20</code>
Desktop screen still blanks / locks despite the setup step	Plasma 6 <code>powerdevil</code> reads <code>~/.config/powerdevilrc</code> — the Plasma-5 <code>powermanagementprofiles.rc</code> is ignored (script $\leq$ v14)	v15+ writes <code>TurnOffDisplayIdleTimeoutSec=-1</code> + <code>AutoSuspendIdleTimeoutSec=-1</code> for all profiles; check System Settings → Power Management
Pairing keeps failing: Failed to set pairable on: <code>org.bluez.Error.Busy</code> in the log, then throwaway bond + "Pairing Not Allowed (0x18)"	THE tile-vs-script root cause (btmon A/B): adapter NOT pairable at <code>pair</code> time → BlueZ answers the pad "No Bonding (0x01)" = throwaway key deleted at the drop; pairable on → "Dedicated Bonding (0x03)" = permanent bond. Pairable wears off (~30 s default timeout / churn after bond removal) and <code>SetPairable</code> can be rejected Busy while the daemon settles	v23+: <code>PairableTimeout=0</code> in <code>main.conf</code> + pairable is SET-AND-VERIFIED in a retry loop right before <code>pair</code> (never assumed). Manual: <code>bluetoothctl pairable on</code> , check <code>bluetoothctl show</code> says <code>Pairable: yes</code> , THEN <code>pair</code>
Bluetooth completely dead after a WARM reboot (Wi-Fi fine)	NXP IW620 combo chip: BT side wedges on warm reboots (<code>hci0: command 0x2039 tx timeout in dmesg</code>); Wi-Fi uses PCIe on the same chip and is unaffected	revive: <code>sudo systemctl stop bluetooth</code> ; <code>sudo modprobe -r btusb</code> ; <code>sleep 3</code> ; <code>sudo modprobe btusb</code> ; <code>sleep 3</code> ; <code>sudo systemctl start bluetooth</code> (v22+ does this automatically). If that fails: COLD boot (power off, re-run the payload) — a warm reboot does not reset the chip
No network at all after boot (LAN and Wi-Fi dead), intermittent on cold AND warm boots	<code>NetworkManager</code> is <code>enabled</code> but the boot transaction sometimes never pulls it in (zero NM lines in the journal)	immediate: <code>sudo systemctl start NetworkManager</code> — back in seconds. permanent (v24+): <code>nm-ensure.service</code> boot watchdog force-starts NM every boot (<code>systemctl status nm-ensure</code>)
Script "does nothing" over one-shot ssh (<code>ssh host "cmd"</code>)	<code>read -p</code> only shows its prompt on a TTY — the script was waiting invisibly (script $\leq$ v19)	v21+ prints every question with <code>echo</code> + answers have timeouts; run interactively (<code>ssh host</code> , then <code>bash ~/ps5-final.sh</code>)

8. Key Paths & One-Liners

Paths

What	Where
Session token	<code>~/.config/ps5-next-session</code>
Session handler log	<code>/tmp/ps5-tty-session.log</code>
Master script	<code>~/ps5-final.sh</code>
Boot cmdline (1080p patch)	<code>/dev/sda2</code> (FAT) → mount, edit <code>cmdline.txt</code>

One-liners

```
# emergency back to gaming
~/rescue-gaming.sh

# which session am I in / what's next
cat ~/.config/ps5-next-session

# clean-slate all sessions (kills SSH!)
sudo loginctl terminate-user steam

# session stack sanity check: exactly one Xorg, one kwin_x11, one plasmashell
pgrep -a Xorg; pgrep -a kwin; pgrep -a plasmashell

# CPU/GPU boost + fan (always fan when boosting)
sudo ~/ps5-linux-tools/ps5_control --fan on
sudo ~/ps5-linux-tools/ps5_control --boost on

# toggle the 1080p boot flag (capture-card fix / revert)
~/toggle-1080p.sh

# re-mount boot partition to edit cmdline manually
sudo mkdir -p /mnt/ps5efi && sudo mount /dev/sda2 /mnt/ps5efi
```

Fan & boost. The PS5's default Linux clocks are conservative; `ps5_control --boost on` raises CPU to 3.5 GHz and GPU to 2.23 GHz. Always enable the fan together with boost — that's what the official PS5 OS does.